# Low-Cost Ocean Instrumentation Workshop

**Woods Hole Oceanographic Institution**

October 19–22, 2026

*Draft agenda — October 9, 2026*

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| **Location** | WHOI DeepTech Facility, Woods Hole, Massachusetts |
| **Hands-on Sessions** | October 23 and October 26–27 (optional) |
| **Supported by** | Schmidt Sciences and the U.S. Office of Naval Research |

## Workshop Objectives

Affordable, scalable ocean observing systems have the potential to substantially expand observations of the coastal and open ocean by enabling measurements at spatial and temporal scales that are difficult to achieve using traditional research-grade instrumentation. This workshop brings together researchers, engineers, and technologists to exchange technical knowledge, demonstrate emerging technologies, discuss common engineering challenges, and develop collaborations that advance reliable, low-cost ocean observing systems.

**Specific objectives:**

- Review recent advances in low-cost oceanographic and marine environmental instrumentation.
- Identify scientific and operational observing needs that can be addressed by affordable sensor technologies.
- Share best practices in sensor design, fabrication, calibration, deployment, and long-term operation.
- Foster collaborations among universities, industry, and nonprofit organizations.
- Develop recommendations and priorities for future technology development and collaborative projects.

## Program

*Talks are 15 minutes plus 5 minutes for questions.*

| **Monday, October 19** | |
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| 08:00–09:00 | *Registration and breakfast* |
| 09:00–09:20 | Welcome and workshop objectives — Tom Farrar (WHOI)<br>Remarks — Emily Shroyer (ONR) |
| 09:20–10:30 | Jessica Sandoval (AquaVela) — Accessible Ocean Technology (AOT) for cost-effective, community-driven ocean observing<br>Andrew Thaler (Oceanography for Everyone) — OpenCTD: a low-cost, open-source CTD for ocean research<br>Kyla Drushka (UW-APL) — Backyard Buoys |
| 10:30–11:00 | *Coffee break* |
| 11:00–12:00 | Kingsley Kojo Darko (University of Ghana) — Low-cost surface drifters for investigating microplastic transport and dispersion in the Gulf of Guinea<br>Wayne Pavalko (Maker Buoy) — Low-cost ocean observing with Maker Buoys<br>Moses Salami (NIOMR, Nigeria; remote) — Challenges and opportunities for affordable ocean observations in the Bight of Benin, Gulf of Guinea<br>Discussion |
| 12:00–13:00 | *Lunch* |

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| 13:00–14:30 | Brennan Phillips (URI) — 3D-printed pressure housings and fiber-optic microtethers for low-cost deep-sea exploration<br>Christopher Piecuch (WHOI) — Observing needs for coastal flooding and sea level<br>Drew Lucas (Scripps) — Waves and tides from a low-cost RTK GNSS buoy<br>Jacob Davis (WHOI) — Design considerations for low-cost, air-deployable wave buoys<br>Discussion |
| 14:30–15:00 | *Coffee break* |
| 15:00–16:30 | Victor Adutwum (KNUST) — A low-cost multi-sensor coastal instrumentation platform for the Keta coast of Ghana<br>Levi Gorrell (WHOI) — Design considerations for a low-cost water-level sensor<br>Jonathan Pfeifer (WHOI) — Open-source sensor engineering and the pHrog low-cost pH sensor<br>Speaker TBD — Waterproof enclosures and performance/cost tradeoffs *[TBC]*<br>Discussion |
| 16:30–17:30 | **Lightning talks (5 minutes each)**<br>Nick Lowell (Lowell Instruments) — From basement to ocean: building practical, affordable instruments<br>Ahmed Aslam Waheed (Maldives National University) — Challenges and opportunities in making ocean observations affordable in the Maldives<br>Michael Edem Donkor (KNUST) — Strengthening local scientific capacity through low-cost instrumentation<br>Akyana Britwum (KNUST) — Research sovereignty and low-cost ocean instrumentation in low-resourced settings<br>Thomas Dwomoh (KNUST) — Monitoring saltwater intrusion in coastal Ghana: developing capabilities with low-cost OhmPi ERT systems |
| **Tuesday, October 20** | |
| 09:00–10:30 | Alex Kinsella (WHOI) — Developing a full-sky thermal camera to advance cloud remote sensing at sea<br>Amina Schartup (Scripps) — A field-deployable method for measuring picomolar mercury levels in natural waters<br>Rupak Loodh (BORI, Bangladesh) — Cost-conscious ocean observations in Bangladesh<br>Oladipo Mumin Olatunji (Kola Daisi University, Nigeria; remote) — Building low-cost scientific instruments for environmental monitoring: a perspective from Nigeria, West Africa<br>Discussion |
| 10:30–11:00 | *Coffee break* |
| 11:00–12:30 | Melissa Omand (URI) — Title TBD<br>Raimund Mullin (Blue Robotics) — Title TBD<br>Zachariah Mears (Blue Robotics) — Modular building blocks: building accessible instrumentation fast with Blue Robotics<br>Oluwabukola Babalola (Miri Africa, Nigeria; remote) — Title TBD<br>Discussion |
| 12:30–13:30 | *Lunch* |
| 13:30–13:50 | Yixuan Song (URI) — Title TBD |
| 13:50–15:00 | **Lightning talks (5 minutes each)**<br>Stefania Emunah (University of Ghana) — Expanding carbon flux observations to coastal African upwelling ecosystems using scalable particle imaging instruments<br>Comfort Opoku (University of Ghana) — Imaging and machine learning approaches to assess zooplankton contributions to deep-ocean carbon export<br>Dipanjan Chaudhuri (UW-APL) — Cost-conscious measurements in the coastal waters of Bangladesh<br>Maurice Edusei (University of Ghana) — From samples to sustained observations: building affordable ocean acidification capacity in Ghana<br>Moise Busogi (CMU-Africa) — Title TBD *[TBC]* |
| 15:00–15:30 | *Coffee break* |

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| 15:30–17:00 | **Demonstrations**<br>Canoe trackers, OMI drifters, uROAM, Snocams, and MINIONs — Melissa Omand (URI)<br>pHrog pH sensor — Jonathan Pfeifer (WHOI)<br>Maker Buoy hardware and kits — Wayne Pavalko (Maker Buoy)<br>Participants are welcome to bring instruments to demonstrate |
| **Wednesday, October 21** | |
| 09:00–10:30 | Shah Selbe (FieldKit) — FieldKit Underwater: development and manufacturing of a low-cost sonde for water-chemistry measurement<br>Evan Heberlein (Cornell) — Water velocity measurements with optical and infrared cameras (tentative)<br>Seth Schweitzer (Cornell) — Measurements of surface flow using infrared and visible-light images<br>Lt. Ekemini Williamson (Nigerian Navy; remote, 5-min lightning talk) — Ocean monitoring at Nigeria's National Hydrographic Agency: efforts and challenges<br>Discussion |
| 10:30–11:00 | *Coffee break* |
| 11:00–11:40 | Bernard Akaawase — Title TBD<br>Onset Computer (HOBO) — Title TBD *[TBC]* |
| 11:40–12:30 | **Open discussion** |
| 12:30–13:30 | *Lunch* |
| 13:30–15:00 | **DeepTech demonstrations and facility tours**<br>1:30 pm: WHOI Rigging Shop tour led by Chris Ross |
| 15:00–15:30 | *Coffee break* |
| 15:30–17:00 | **Breakout discussions** |
| **Thursday, October 22** | |
| 09:00–10:30 | **Breakout reports** |
| 10:30–10:45 | *Coffee break* |
| 10:45–11:45 | **Collaborative opportunities and workshop roadmap** |
| 11:45–12:30 | **Workshop report planning and closing remarks** |
| 12:30–13:30 | *Lunch; main workshop adjourns* |
| 13:30–16:00 | **Optional: OpenCTD build preparation — introduction to soldering and software setup (WHOI DeepTech facility)**<br>Andrew Thaler (Oceanography for Everyone) |
| **Friday, October 23 and Monday–Tuesday, October 26–27 (optional)** | |
| All day | **Hands-on development sessions — WHOI DeepTech facility**<br>OpenCTD build in small teams, led by Andrew Thaler<br>Short presentation by Samuel Tawiah (University of Ghana): Reducing the cost of ocean observations in Ghana: tools, instrumentation, measurements, and maintenance<br>Other builds and test deployments to be announced |